

Lecture 10 (February 24), continued

We will move onto a new chapter on partially ordered sets.

The goal: Sperner's Theorem

We will try to build up to the following theorem.

Theorem 1 (Sperner (1928)).

Let $S = \{I_1, \dots, I_N\}$ be different subsets of $[n]$ such that for all $i \neq j$, I_i and I_j are *incomparable* (i.e. $I_i \not\subseteq I_j$ and $I_j \not\subseteq I_i$). Then $N \leq \binom{N}{\lfloor n/2 \rfloor}$.

(Equality occurs when S comprises all subsets of cardinality $\lfloor n/2 \rfloor$, or, symmetrically, all subsets of cardinality $\lceil n/2 \rceil$.)

We can find a generalization of this theorem that is easier to prove due to induction. We need to introduce the notion of a **poset** (short for partially ordered set).

Poset definitions

Definition (Posets).

A **poset** is a set P with a binary relation $\cdot \leq \cdot$, with the following axioms:

1. For all elements $a \in P$, $a \leq a$ (reflexivity)
2. If $a \leq b$ and $b \leq a$, that implies $a = b$.
3. If $a \leq b$ and $b \leq c$, then $a \leq c$ (transitivity).

We also introduce some more notation:

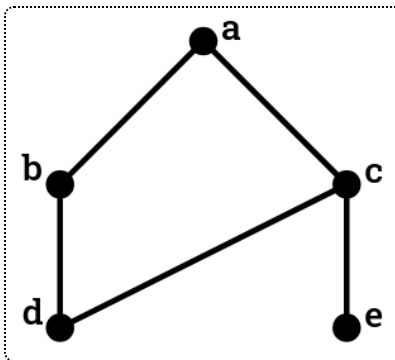
- $a < b$ means $a \leq b$ and $a \neq b$

- $a \lessdot b$ (" a is **covered** by b " or " b **covers** a ") means $a < b$ and $\nexists c \in P$ such that $a < c < b$. (There is no element between a and b .)

Can you reconstruct a covering relation from a poset? Not necessarily: consider the poset $\mathbb{R}$ with the typical $\leq$ (actually a completely ordered set, but that's a special case). There are no covering relations at all so you can't recover anything.

However, it turns out that if your poset is finite, you can recover it from covering relations. As such, we can represent finite posets with **Hasse diagrams**, which are essentially graphs where covering relations are edges.

Although these graphs are directed graph, the direction is usually denoted by verticality (so that the larger element is above the smaller one); this enforces transitivity. For example, in the following Hasse diagram, we have $b \lessdot a$, $c \lessdot a$, $d \lessdot b$, $d \lessdot c$, and $e \lessdot c$.



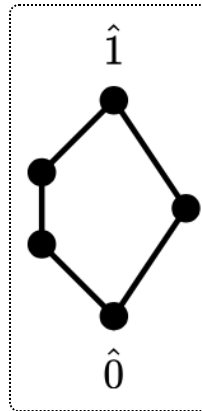
a is the unique maximal element in this poset (which may not exist in general), so we can also denote it $\hat{1}$. A unique minimal element would be denoted $\hat{0}$.

More definitions:

- a, b are **incomparable** if $a \not\leq b$ and $b \not\leq a$. (Otherwise, they are comparable.)
- A **chain** $C \subset P$ is a subset such that any two elements of C are comparable.
- A **saturated chain** C can be ordered as $C = \{c_1 \lessdot c_2 \lessdot \dots \lessdot c_k\}$.
- An **anti-chain** $A \subset P$ is a subset where any two elements are incomparable.
- A poset P is **ranked** if there exists a "rank function" $\rho: P \rightarrow \mathbb{Z}$ such that for any pairs $a \lessdot b$, $\rho(a) + 1 = \rho(b)$. By convention, the image of ρ for a finite ranked poset is $\{0, 1, \dots, \ell\}$ (in theory you can add any constant).

In the above example, b and e are incomparable, $\{d, b, a\}$ is a saturated chain, and $\{b, c\}$ is an anti-chain. It is a ranked poset: $\rho(d) = \rho(e) = 0$, $\rho(b) = \rho(c) = 1$, $\rho(a) = 2$, with the maximal rank ℓ being 2.

Note that not all finite posets are ranked! For example, see the following:

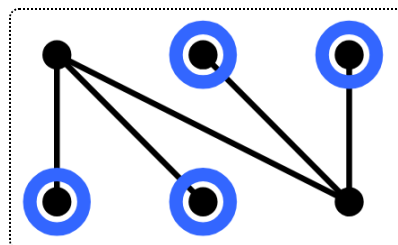


For a ranked poset P , let $P_i = \{a \in P \mid \rho(a) = i\}$. The "rank numbers" r_i are the cardinalities of each P_i . In our original example, $r_0 = 2$, $r_1 = 2$, and $r_2 = 1$.

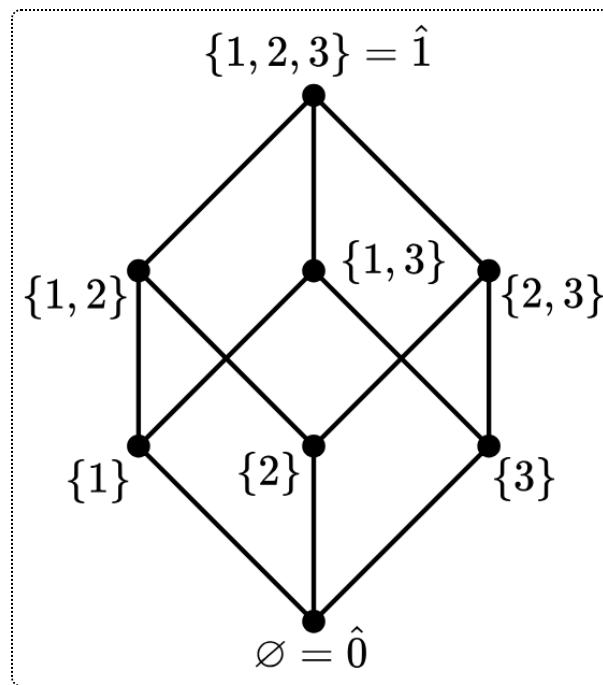
A finite ranked poset is called:

- **rank-symmetric** if $r_i = r_{\ell-i}$ for all $i = 0, \dots, \ell$.
- (rank-) **unimodal** if $r_0 \leq r_1 \leq \dots \leq r_s \geq r_{s+1} \geq \dots \geq r_\ell$. (The rank numbers monotonically increase and then monotonically decrease.)
- **Sperner** if the maximal size of an anti-chain equals $\max(r_0, r_1, \dots, r_\ell)$.

You can certainly find an anti-chain of rank $\max(r_0, r_1, \dots, r_\ell)$: suppose the maximum is r_k ; then take all elements of rank k . It just may not be maximal: the following poset has maximal rank number 3 but a circled anti-chain of 4 elements.



Sperner's Theorem is equivalent to saying that the **Boolean lattice** B_n , the poset of all subsets $I \subset [n]$ ordered by $\subseteq$, is Sperner. Here's what B_3 looks like:



It is not difficult to show that B_n is rank-symmetric and unimodal, since the rank numbers are $r_i = \binom{n}{i}$.

In the next lecture, we will work on actually proving the theorem.

Lecture 11 (February 26)

Recall the poset-related definitions and terminology from last time.

Proof of Sperner's Theorem

The following definition is crucial.

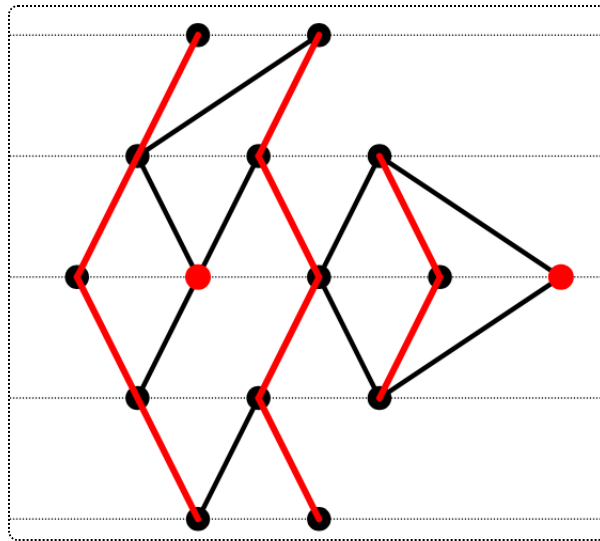
Definition (Symmetric chain decompositions).

A **symmetric chain decomposition** (SCD) of a poset P is a decomposition into a disjoint union of saturated chains

$$P = c_1 \sqcup c_2 \sqcup \cdots \sqcup c_m$$

such that for all i , c_i starts at rank j and ends at rank $l - j$.

For example, the red paths/dots in the below diagram represent chains in a symmetric chain decomposition of the poset.



Lemma (SCD implies properties).

If P has a SCD, then P is rank-symmetric, unimodal, and Sperner.

Proof of Lemma. Rank-symmetry and unimodality are not hard to see by counting chains in our decomposition that start/end at each rank. To show that P is Sperner, note that any anti-chain A intersects each chain c_i at most once, so $|A|$ is at most the number m of chains in our SCD, which is just $r_{\lfloor \ell/2 \rfloor}$, i.e. the maximum rank number, as desired. (Remember that ℓ is the maximum rank.)

As a result, the following theorem implies Sperner's Theorem.

Theorem 2 (de Bruijn (1948)).

B_n (the Boolean lattice) has a symmetric chain decomposition.

This theorem has a few immediate corollaries. $\binom{[n]}{k}$ denotes the set of k -element subsets of $[n]$, and an order-preserving map f in a poset satisfies $a \leq f(a)$ for all a .

Corollary 3.

There exists an order-preserving injective map $f: \binom{[n]}{i} \rightarrow \binom{[n]}{i+1}$ if $i < \frac{n}{2}$.

Corollary 4.

There exists an order-preserving bijection $f: \binom{[n]}{i} \rightarrow \binom{[n]}{n-i}$ if $i < \frac{n}{2}$.

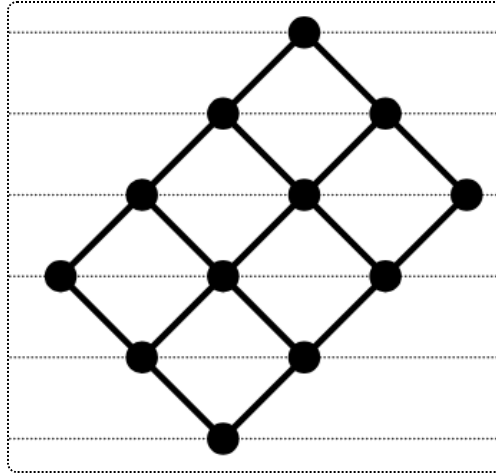
Proof. We will need the notion of a "product of posets" $P \times Q$:

$$P \times Q := \{(p, q) \mid p \in P, q \in Q\}$$

$$(p, q) \leq (p', q') \iff p \leq_P p' \quad \text{and} \quad q \leq_Q q'$$

Equivalently, $(p, q) \triangleleft (p', q')$ where $q \triangleleft q'$ and $(p, q) \triangleleft (p', q)$ when $p \triangleleft p'$.

For example, consider the **n-chain** poset $[n] = \{1 \triangleleft 2 \triangleleft \dots \triangleleft n\}$. Its Hasse diagram is just a vertical line. Then $[3] \times [4]$ is the grid below.



Observe that the Boolean lattice B_n is just

$$\underbrace{[2] \times [2] \times [2] \times \dots \times [2]}_{n \text{ times}},$$

each factor corresponding to one element of the set $[n]$.

So, it suffices to prove the following general claim.

Claim (Products of chains have SCDs).

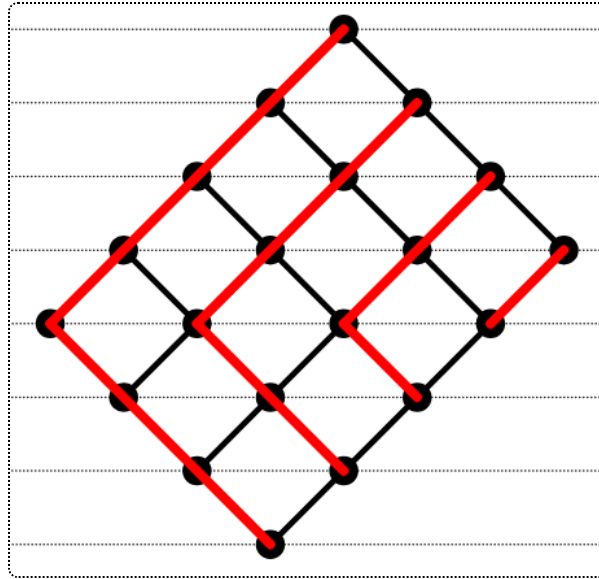
Any product of chains $[n_1] \times [n_2] \times \dots \times [n_k]$ has a SCD.

(Although this is more general, it is easier to prove since if we use induction, we can assume a stronger inductive hypothesis.)

Lemma 5 (Products of two chains).

$[m] \times [n]$ has a SCD.

"Proof" of Lemma. Emulate the following example for $[4] \times [5]$:



We can verify that each of these chains are indeed symmetric.

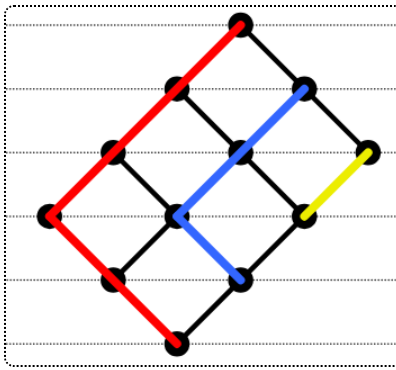
Lemma 6 (Products of SCDs).

If P and Q have SCDs, then $P \times Q$ has a SCD.

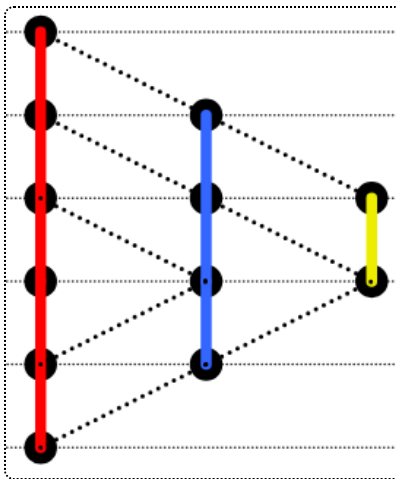
Proof of Lemma. Decompose $P = c_1 \sqcup c_2 \sqcup \dots \sqcup c_a$ and $Q = c'_1 \sqcup c'_2 \sqcup \dots \sqcup c'_b$. Then pick a SCD for each product $c_i \times c'_j$ (which is a product of two chains); combining them yields a SCD of $P \times Q$.

This finishes the theorem.

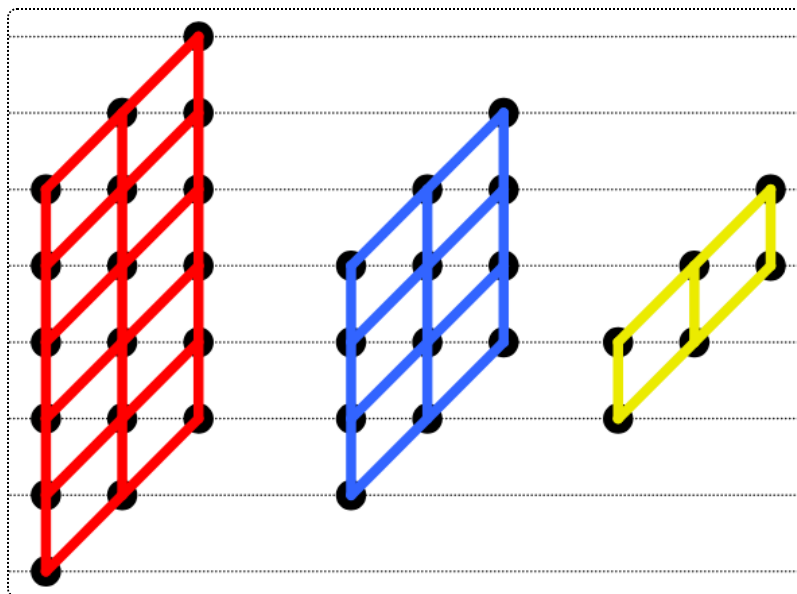
This may still seem unintuitive, so here's an example. Consider $([4] \times [3]) \times [3]$. We have three chains in the SCD of $[4] \times [3]$; color them red, blue, and yellow.



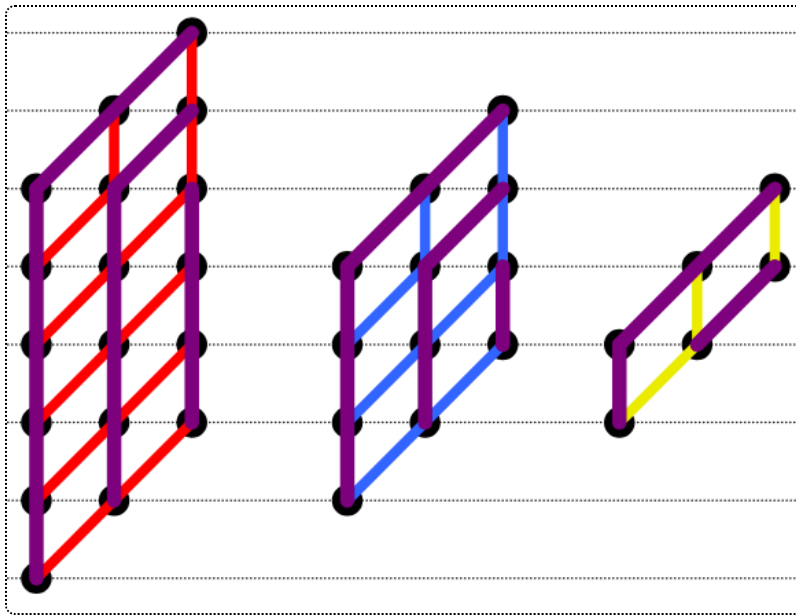
We can straighten out the chains. Note that we don't actually need care about the covering relations between the different chains now, since we won't use them to construct our SCD, so we will remove them after this diagram.



Now, when we multiply by $[3]$, each chain becomes a grid.



Note that each grid is still symmetric about the median rank, so it is indeed valid to combine SCDs for each individual grid, shown in purple:



As such, the Boolean lattice has a SCD, and therefore it is Sperner.

More theorems

We will move onto other poset-related topics in future classes.

Theorem 7 (Dilworth (1950)).

The maximum cardinality of an anti-chain in P equals the minimum number of chains needed to cover all elements in P .

Note that the $\leq$ direction is easy, since any chain and any anti-chain can intersect in at most one element.

The "dual" of Dilworth's Theorem is also true.

Theorem 8 (Mirsky (1971)).

The maximum cardinality of a chain in P equals the minimal number anti-chains needed to cover all elements in P .